Ansible ANWENDER

11. März 2026 | Dortmund

9. Ansible Anwendertreffen

Powered by Red Hat and MATERNA

Logging Ansible Output

Overview and demo of different methods

Computacenter

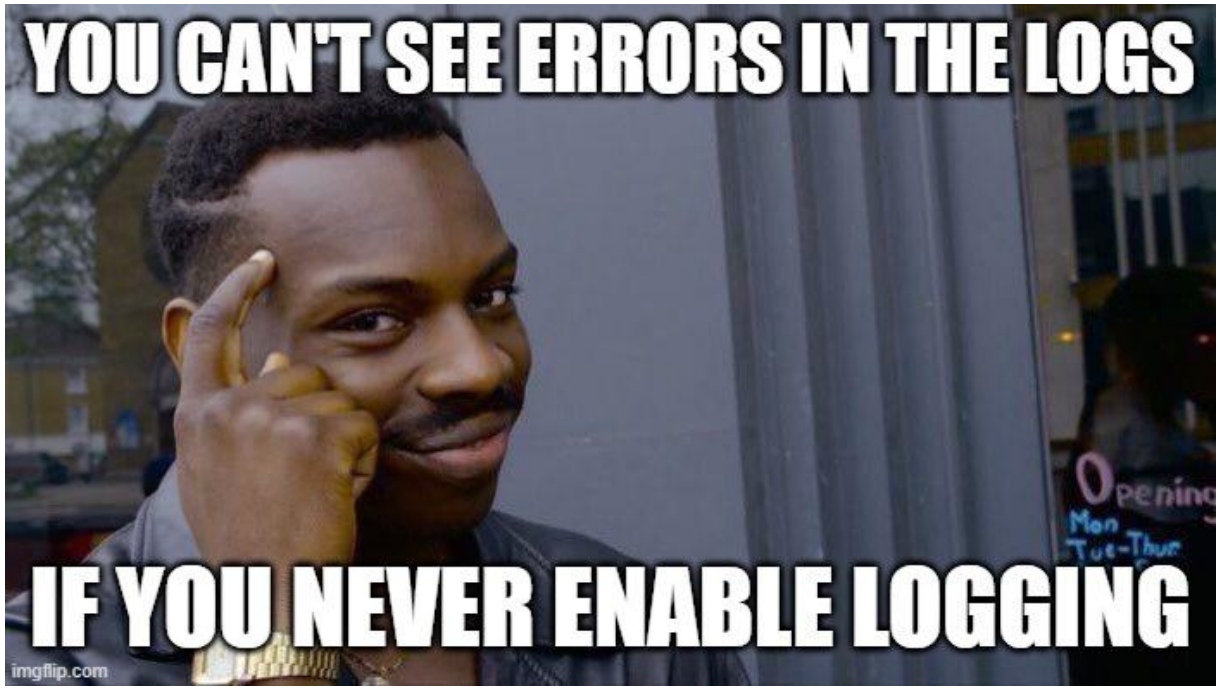
Logging Ansible Output

Agenda

- Who needs logs anyways?
- Logging Ansible on CLI
- Professional & specialised utilities
- Recap

Ansible Logging

Not enabled by default! Output goes to STDOUT



Logging for ansible-core

Enable in Ansible configuration file

Save to single log file on the control node

```
[defaults]
log_path = ~/.ansible.log
```

- Path to Log-File must exist (!) and be writeable for the user

```
$ ansible-playbook demo.yml
[WARNING]: log file at /etc/ansible/ansible.log is not writeable and we cannot create it, aborting

PLAY [Demo play] ***********************************************************************************

TASK [Gathering Facts] *****************************************************************************
ok: [localhost]
```

Additionally, send logs to external systems with callback plugins

- community.general.splunk
- community.general.elastic
- community.general.opentelemetry
- …

Save to multiple log files on the control node

Use callback (notification) plugin “community.general.log_plays”

```
[defaults]
callbacks_enabled = community.general.log_plays

[callback_log_plays]
log_folder = logs/hosts
```

- creates files like “logs/hosts/node1.example.com”

```
Feb 28 2025 11:43:10 - package-installation.yml - Gathering Facts - gather_facts - OK - omitted

Feb 28 2025 11:43:14 - package-installation.yml - Install tmux - ansible.builtin.package - OK - {"module_args": {"name": ["tmux"],

Feb 28 2025 11:43:22 - package-installation.yml - Gathering Facts - gather_facts - OK - omitted

Feb 28 2025 11:43:25 - package-installation.yml - Install tmux - ansible.builtin.package - OK - {"module_args": {"name": ["tmux"],
```

What about logging on the managed nodes?

- Ansible writes stuff to syslog (/var/log/messages)

Logging for ansible-core

Without adjusting Ansible configuration file

Save output in multiple log files on the control node

- Requires some trickery with Ansible environment variables!

```
$ ANSIBLE_LOG_PATH=playbook.$(date +%Y%m%d-%HH%MM%SS).log ansible-playbook demo.yml

PLAY [Demo play] ***********************************************************************

TASK [Gathering Facts] *****************************************************************
ok: [localhost]
```

- Path to Log-File **must** exist (!) and be writeable for the user
- Creates a new file with timestamp on every playbook run
- in example above in the same folder, should be git-ignored
- **TIP**: Add a bash alias (e.g. to ~/.bash_aliases) for ansible-playbook itself

```
alias ansible-playbook='ANSIBLE_LOG_PATH=playbook.$(date +%Y%m%d-%HH%MM%SS).log ansible-playbook'
```

```
2025-02-27 16:26:44,616 p=130493 u=timgrt n=ansible | PLAY [Create Workshop environment] ****************************
2025-02-27 16:26:44,620 p=130493 u=timgrt n=ansible | TASK [Gathering Facts] ****************************************
2025-02-27 16:26:46,132 p=130493 u=timgrt n=ansible | ok: [localhost]
2025-02-27 16:26:46,142 p=130493 u=timgrt n=ansible | TASK [Get package facts] *************************************
2025-02-27 16:26:47,169 p=130493 u=timgrt n=ansible | ok: [localhost]
2025-02-27 16:26:47,182 p=130493 u=timgrt n=ansible | TASK [Install Podman] ****************************************
2025-02-27 16:26:47,198 p=130493 u=timgrt n=ansible | skipping: [localhost]
2025-02-27 16:26:47,212 p=130493 u=timgrt n=ansible | TASK [Pull image for managed node containers] ******************
2025-02-27 16:26:47,779 p=130493 u=timgrt n=ansible | ok: [localhost]
2025-02-27 16:26:47,792 p=130493 u=timgrt n=ansible | TASK [Start managed node containers, publish 3 ports for each co
2025-02-27 16:26:49,064 p=130493 u=timgrt n=ansible | changed: [localhost] => (item=node1)
2025-02-27 16:26:50,255 p=130493 u=timgrt n=ansible | changed: [localhost] => (item=node2)
2025-02-27 16:26:51,494 p=130493 u=timgrt n=ansible | changed: [localhost] => (item=node3)
2025-02-27 16:26:51,510 p=130493 u=timgrt n=ansible | TASK [Create folder for workshop inventory] ******************
2025-02-27 16:26:51,723 p=130493 u=timgrt n=ansible | changed: [localhost]
2025-02-27 16:26:51,738 p=130493 u=timgrt n=ansible | TASK [Create workshop inventory file] ************************
2025-02-27 16:26:52,139 p=130493 u=timgrt n=ansible | changed: [localhost]
2025-02-27 16:26:52,153 p=130493 u=timgrt n=ansible | TASK [Add block to ssh_config for easy SSH access to managed nod
2025-02-27 16:26:52,462 p=130493 u=timgrt n=ansible | changed: [localhost]
2025-02-27 16:26:52,479 p=130493 u=timgrt n=ansible | TASK [Deploy ansible.cfg to home directory] ******************
2025-02-27 16:26:52,775 p=130493 u=timgrt n=ansible | changed: [localhost]
2025-02-27 16:26:52,790 p=130493 u=timgrt n=ansible | TASK [Backup file of .ansible.cfg created] *******************
2025-02-27 16:26:52,807 p=130493 u=timgrt n=ansible | skipping: [localhost]
2025-02-27 16:26:52,821 p=130493 u=timgrt n=ansible | TASK [Check for existing SSH keypair] ************************
2025-02-27 16:26:53,070 p=130493 u=timgrt n=ansible | ok: [localhost]
2025-02-27 16:26:53,106 p=130493 u=timgrt n=ansible | TASK [Check if OpenSSH keypair does not match target configurati
2025-02-27 16:26:53,115 p=130493 u=timgrt n=ansible | skipping: [localhost]
2025-02-27 16:26:53,130 p=130493 u=timgrt n=ansible | TASK [Abort playbook if keypair was found and does not match tar
2025-02-27 16:26:53,140 p=130493 u=timgrt n=ansible | skipping: [localhost]
2025-02-27 16:26:53,156 p=130493 u=timgrt n=ansible | TASK [Create OpenSSH keypair for accessing managed nodes] *****
2025-02-27 16:26:54,819 p=130493 u=timgrt n=ansible | changed: [localhost]
2025-02-27 16:26:54,877 p=130493 u=timgrt n=ansible | PLAY [Prepare managed nodes] ********************************
2025-02-27 16:26:54,880 p=130493 u=timgrt n=ansible | TASK [Gathering Facts] ****************************************
2025-02-27 16:26:57,478 p=130493 u=timgrt n=ansible | ok: [node2]
2025-02-27 16:26:57,599 p=130493 u=timgrt n=ansible | ok: [node3]
2025-02-27 16:26:57,744 p=130493 u=timgrt n=ansible | ok: [node1]
2025-02-27 16:26:57,767 p=130493 u=timgrt n=ansible | TASK [Install SSH daemon] ************************************
2025-02-27 16:27:39,968 p=130493 u=timgrt n=ansible | changed: [node1]
2025-02-27 16:27:41,405 p=130493 u=timgrt n=ansible | changed: [node3]
2025-02-27 16:27:41,715 p=130493 u=timgrt n=ansible | changed: [node2]
2025-02-27 16:27:41,721 p=130493 u=timgrt n=ansible | TASK [Start SSH daemon] **************************************
2025-02-27 16:27:44,239 p=130493 u=timgrt n=ansible | changed: [node1]
2025-02-27 16:27:44,298 p=130493 u=timgrt n=ansible | changed: [node2]
2025-02-27 16:27:44,421 p=130493 u=timgrt n=ansible | changed: [node3]
2025-02-27 16:27:44,425 p=130493 u=timgrt n=ansible | TASK [Remove /run/nologin to be able to login as unprivileged us
2025-02-27 16:27:46,690 p=130493 u=timgrt n=ansible | ok: [node3]
2025-02-27 16:27:46,918 p=130493 u=timgrt n=ansible | ok: [node2]
2025-02-27 16:27:46,967 p=130493 u=timgrt n=ansible | ok: [node1]
2025-02-27 16:27:46,971 p=130493 u=timgrt n=ansible | TASK [Add public key of workshop SSH keypair to authorized_keys
2025-02-27 16:27:49,269 p=130493 u=timgrt n=ansible | changed: [node1]
2025-02-27 16:27:49,313 p=130493 u=timgrt n=ansible | changed: [node2]
2025-02-27 16:27:49,587 p=130493 u=timgrt n=ansible | changed: [node3]
2025-02-27 16:27:49,639 p=130493 u=timgrt n=ansible | PLAY RECAP ***************************************************
2025-02-27 16:27:49,639 p=130493 u=timgrt n=ansible | localhost                  : ok=10   changed=6    unreachable=0
2025-02-27 16:27:49,639 p=130493 u=timgrt n=ansible | node1                      : ok=5    changed=3    unreachable=0
2025-02-27 16:27:49,639 p=130493 u=timgrt n=ansible | node2                      : ok=5    changed=3    unreachable=0
2025-02-27 16:27:49,639 p=130493 u=timgrt n=ansible | node3                      : ok=5    changed=3    unreachable=0
```

Logging for ansible-navigator

CLI and TUI for running Ansible content

- Creates multiple log files and is highly customizable
- creates folder for log files itself (if it has sufficient permissions)
- (Playbook-)Logs are stored as JSON
  - Use replay function to show output again
  - Use interactive mode to drill into single tasks

Ansible Navigator is the swiss army knife for local Ansible execution

- can use Execution Environment container or the local Ansible installation
- all Ansible binaries (e.g. ansible-doc, ansible-inventory, …) in one utility
- stdout mode or interactive TUI mode

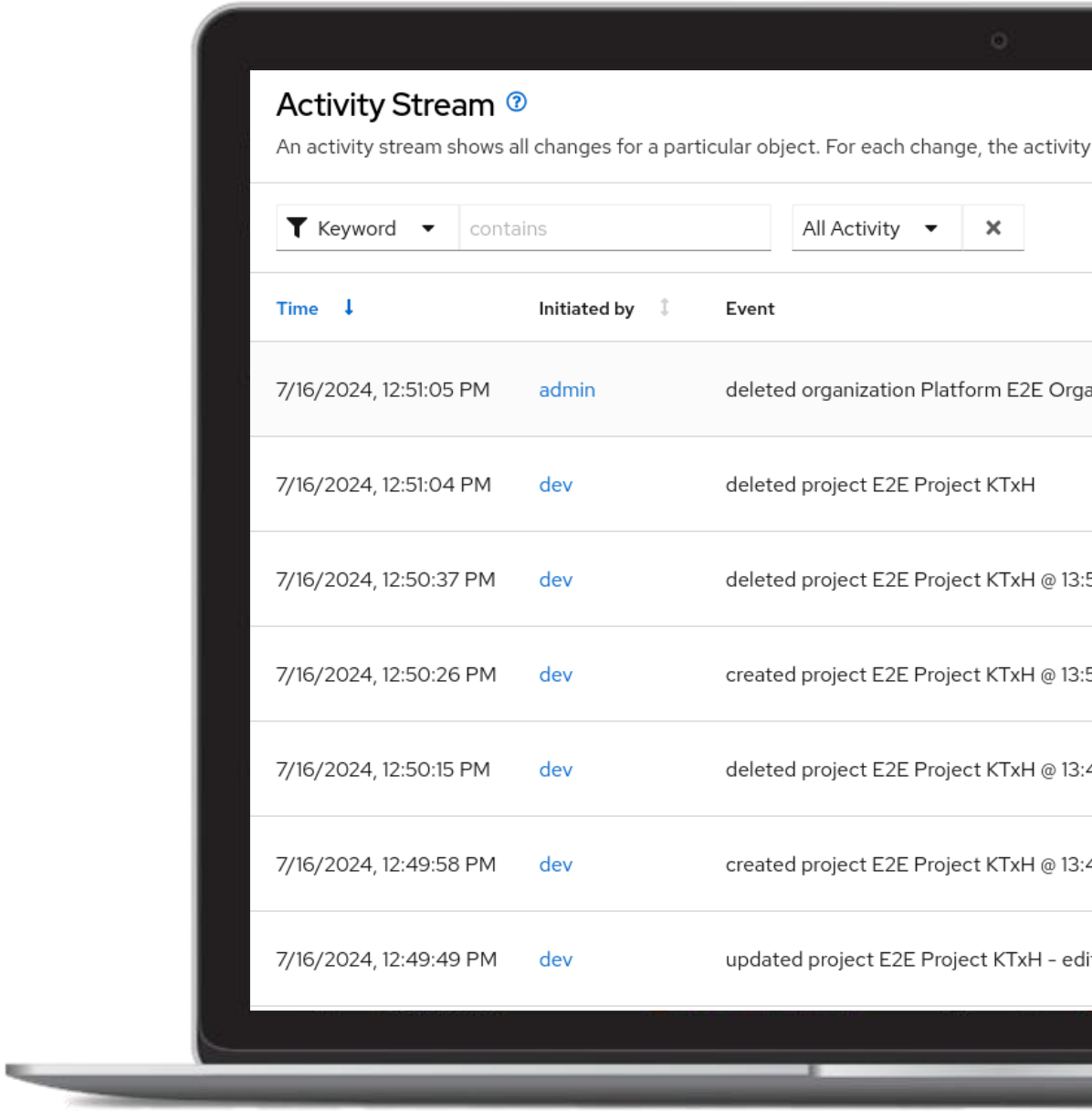
```yaml
---
ansible-navigator:
  execution-environment:
    enabled: false
    image: ghcr.io/ansible-community/community-ee-base:latest
    pull:
      policy: missing
  logging:
    level: warning
    file: logs/ansible-navigator.log
  mode: stdout
  playbook-artifact:
    enable: true
    save-as: "logs/{playbook_status}-{playbook_name}-{time_stamp}.json"
```

Ansible Automation Platform

Logging and Auditing builtin

- stores Job Logs (Playbook runs) in Postgres database
  - who executed the template
  - what was the result of play or every single task
  - ...
- also stores changes to all AAP objects (Activity Stream)
  - e.g. who modified the Job template

- can **send detailed logs to third-party external log aggregation service**
  - easier to setup than locally
  - Splunk
  - Loggly
  - Sumologic
  - Elastic Stack (formerly ELK stack)
  - Custom



ARA

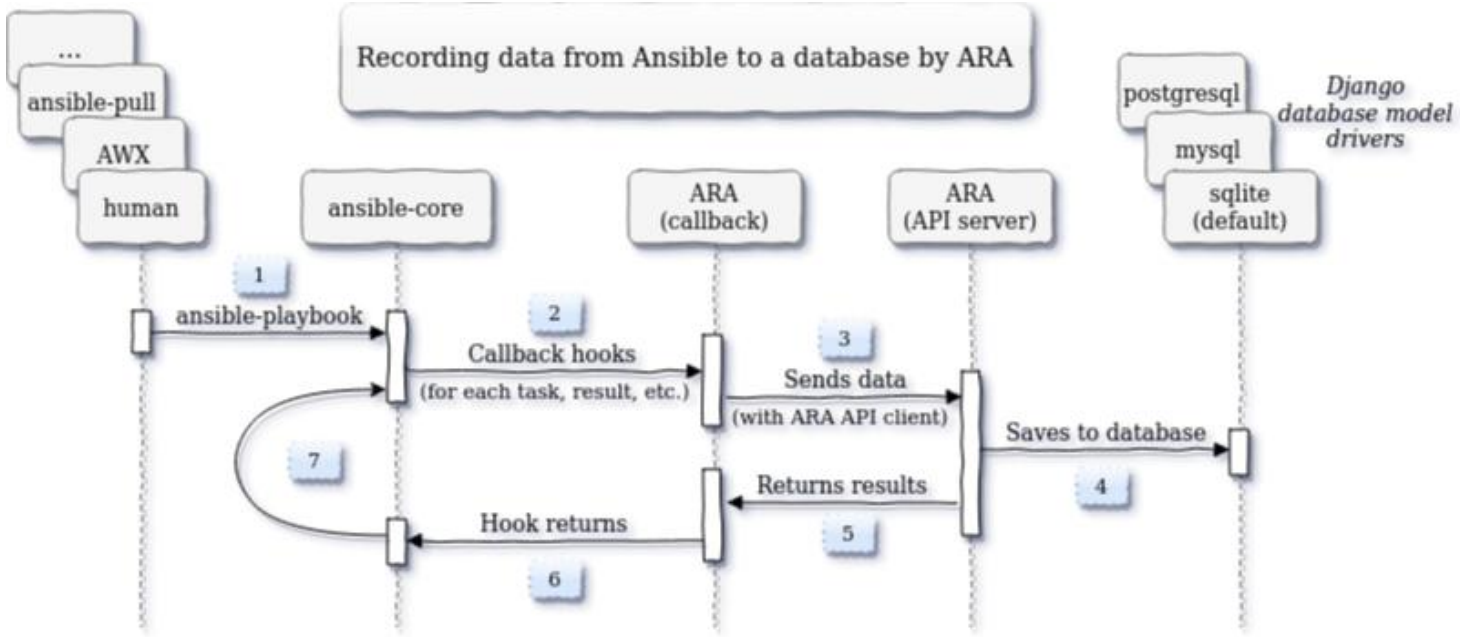
Highly specialised logging tool for Ansible

ARA – Ansible Records Ansible

provides Ansible reporting by recording ansible and ansible-playbook commands regardless of how and where they run.

- Writes results to SQLite, MySQL or PostgreSQL databases via custom **callback plugin**.
- Any recent Linux distribution or Mac OS with Python >=3.8
- Supports authentication
- The ara package (containing the ARA Ansible plugins) must be installed for the same python interpreter as Ansible itself

- **Logs can be read via CLI, API or UI**



Logging Ansible Output

Recap

Should you log Ansible output?

Yes, no, maybe, depends, …

- Depends on your Use-Case and your (customers) requirements!
  - **But**, having logs is never bad…
  - Always nice to be able to „prove“ what Ansible did, how it ended and who did it

- Multiple options for logging Ansible output
- External logging systems can consume Ansible logs
  - Ansible can send logs to external systems during runtime

Type	Implementation Difficulty	Usability	Readability
Single log file on Controller (builtin)	easy	easy	medium
Multiple log files on Controller (builtin)	medium	easy	easy
Multiple log files on Controller (Collection)	easy	easy	medium
Log file on Managed Nodes	easy	medium	medium to hard
Navigator	easy to medium	easy	easy to medium
ARA	medium to hard	easy to medium	easy to medium
AAP	easy to medium	easy	easy

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